

Course Pack 02

Dirac Notation and Matrix Groups
Quantum Entanglement & Quantum Computing (UE23EC343AB3)

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What Is In This Pack

Part I: Dirac notation. Linear algebra in a new outfit. One genuinely new object shows up, the **outer product**, and everything interesting is built from it.

Part II: matrix groups. Starts from scratch with what a **group** actually is, then swaps the abstract moves for matrices. We end up with $U(n)$, the subgroup $SU(n)$ hiding inside it, and a one-line determinant test that sorts every gate you know.

Not here on purpose: dual spaces, the spectral theorem, Lie algebras, global phase, the measurement postulate. Each of these turns up later, when there is something to do with it.

How to use this: every topic ends with four short questions. Do them before moving on. They take two minutes and they are the difference between reading notes and knowing things.

On the Notation

Paul Dirac took the word *bracket*, sliced it in half, and named the two pieces *bra* and *ket*. Physicists have been living with this decision since 1939. Annoyingly, it was a good one.

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Part I

Dirac Notation

1 The Dictionary

Dirac notation is mostly **linear algebra wearing quantum-mechanics clothing**.

Nothing here is new mathematics. We are just changing the notation so that we can write quantum states and operators without carrying around giant column vectors everywhere.

Linear algebra	Dirac notation
column vector v	ket $ v\rangle$
conjugate transpose $v^\dagger$	bra $\langle v $
inner product $u^\dagger v$	$\langle u v\rangle$
matrix A times vector v	$A v\rangle$
$u^\dagger A v$	$\langle u A v\rangle$
$v u^\dagger$	outer product $ v\rangle\langle u $
standard basis e_1, e_2	$ 0\rangle, 1\rangle$

The Big Picture

There are really only two new words to keep straight:

ket = vector

bra = conjugate transpose of that vector

Everything else is built from these.

So don't memorise Dirac notation as a collection of mysterious brackets. Translate it back to linear algebra whenever you get stuck.

Ket and Bra

For $|v\rangle \in \mathbb{C}^n$,

$$|v\rangle = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}, \quad \langle v| = |v\rangle^\dagger = (\overline{v_1} \quad \cdots \quad \overline{v_n}).$$

So:

ket = column

bra = conjugate-transposed row

That's the whole definition.

A Bra Is Basically a Machine

A ket is a vector.

A bra takes a vector and gives you a number:

$$\langle u| : |v\rangle \longmapsto \langle u|v\rangle \in \mathbb{C}.$$

For example,

$$|u\rangle = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \implies \langle u| = (u_1^* \quad u_2^*).$$

Then

$$\langle u|v\rangle = (u_1^* \quad u_2^*) \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = u_1^* v_1 + u_2^* v_2.$$

So the bra is not some mysterious new quantum object. It is just the familiar conjugate-transposed vector doing its job.

The Computational Basis

For a single qubit, the standard basis vectors are written as

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Their corresponding bras are

$$\langle 0| = (1 \quad 0), \quad \langle 1| = (0 \quad 1).$$

So, for example,

$$\langle 0|1\rangle = (1 \quad 0) \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 0.$$

Nothing mysterious happened.

We just took the usual inner product of two orthogonal basis vectors.

One Remark, Then Move On

There is a more careful mathematical story involving *dual spaces*. It matters in the full mathematical treatment, but it is not needed for what we are doing here.

For this course, the useful picture is simply:

$$|\psi\rangle \xrightarrow{\dagger} \langle\psi|$$

and

$$\langle\phi||\psi\rangle = \text{a number}$$

That's enough for now. We'll bring in the heavier machinery only when we actually

need it.

Quick Practice: The Dictionary

Q1. Given

$$|\psi\rangle = \begin{pmatrix} 1 \\ i \end{pmatrix},$$

write $\langle\psi|$.

Q2. Given

$$\langle\phi| = (2 \quad -i),$$

write $|\phi\rangle$.

Q3. Compute $\langle 0|1\rangle$ and $\langle 1|0\rangle$.

Q4. Given

$$|\psi\rangle = \begin{pmatrix} a \\ b \end{pmatrix},$$

write $\langle\psi|$, then compute $\langle\psi|\psi\rangle$.

Answers: The Dictionary

Q1. Transpose and conjugate. The i becomes $-i$:

$$\langle\psi| = (1 \quad -i).$$

Q2. Same move backwards:

$$|\phi\rangle = \begin{pmatrix} 2 \\ i \end{pmatrix}.$$

Q3. Both are zero. The basis is orthogonal, and it does not matter which way round you take the inner product here because 0 is its own conjugate.

Q4. Conjugate each entry, then multiply:

$$\langle\psi| = (\bar{a} \quad \bar{b}), \quad \langle\psi|\psi\rangle = \bar{a}a + \bar{b}b = |a|^2 + |b|^2.$$

Note this is always a real number ≥ 0 , which is exactly why $\langle\psi|\psi\rangle$ can be used as a length. If you got $a^2 + b^2$, you forgot the conjugate.

2 Numbers and Operators

Now we get to the first genuinely useful new idea.

Put a bra and a ket together in two different orders:

$$\boxed{\langle\phi| |\psi\rangle} \quad \text{vs.} \quad \boxed{|\psi\rangle \langle\phi|}$$

Same ingredients. **Completely different objects.**

The fastest way to tell which one you have is to check the dimensions.

Expression	Shape	Result
$\langle\phi \psi\rangle$	$(1 \times n)(n \times 1)$	number
$ \psi\rangle\langle\phi $	$(n \times 1)(1 \times n)$	operator

Order Is Everything

The order is not decoration.

$$\langle\phi|\psi\rangle = \text{scalar}$$

but

$$|\psi\rangle\langle\phi| = \text{operator}$$

When confused, do not guess.

Count the dimensions.

$$(1 \times n)(n \times 1) = 1 \times 1$$

gives a number, while

$$(n \times 1)(1 \times n) = n \times n$$

gives a matrix.

2.1 Inner Product vs. Outer Product

Same Vectors, Different Order

Take the computational basis

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Inner product:

$$\langle 0|1\rangle = (1 \ 0) \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 0.$$

A number. We asked:

“How much of $|1\rangle$ points along $|0\rangle$?”

Answer: none.

Outer product:

$$|0\rangle\langle 1| = \begin{pmatrix} 1 \\ 0 \end{pmatrix} (0 \ 1) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Now we have a matrix, so this thing can *act* on a state.

What Does the Outer Product Actually Do?

Do not just stare at

$$|0\rangle \langle 1|.$$

Make it act on something.

Apply it to $|1\rangle$:

$$|0\rangle \langle 1| |1\rangle = |0\rangle \langle 1|1\rangle = |0\rangle.$$

Apply it to $|0\rangle$:

$$|0\rangle \langle 1| |0\rangle = |0\rangle \langle 1|0\rangle = 0.$$

Therefore,

$$|0\rangle \langle 1| : |1\rangle \mapsto |0\rangle, \quad |0\rangle \mapsto 0$$

So $|0\rangle \langle 1|$ is an operator that:

1. looks for the $|1\rangle$ component;
2. extracts it using $\langle 1|$;
3. sends that component into the $|0\rangle$ direction.

That is the useful mental picture:

$$|\psi\rangle \langle \phi| = \text{“take the } |\phi\rangle \text{ component and send it toward } |\psi\rangle\text{”}$$

Read Operators Right-to-Left

For

$$|\psi\rangle \langle \phi| |v\rangle,$$

start from the right:

$$|v\rangle \longrightarrow \langle \phi|v\rangle \longrightarrow |\psi\rangle \langle \phi|v\rangle.$$

The bra first extracts a number. The ket then tells you the direction in which that number lives.

So:

$$|\psi\rangle \langle \phi| |v\rangle = \langle \phi|v\rangle |\psi\rangle$$

This is one of the most useful ways to read Dirac notation: **don't just ask what the symbols are; ask what the operator is doing.**

2.2 The Regrouping Trick

There is one piece of algebra hiding underneath all of this.

Matrix multiplication is associative:

$$(AB)C = A(BC).$$

So

$$(|\psi\rangle \langle\phi|) |v\rangle = |\psi\rangle (\langle\phi| |v\rangle).$$

Using Dirac notation,

$$\boxed{(|\psi\rangle \langle\phi|) |v\rangle = |\psi\rangle \langle\phi|v\rangle}$$

The middle pair,

$$\langle\phi| |v\rangle,$$

collapses to a number.

That is why the expression becomes

$$(\text{number}) |\psi\rangle.$$

The Collapse Pattern

Whenever you see

$$|a\rangle \langle b| |c\rangle,$$

look at the middle:

$$\langle b| |c\rangle.$$

It is an inner product, so it becomes a number:

$$|a\rangle \underbrace{\langle b|c\rangle}_{\text{number}}.$$

In other words:

$$\boxed{\text{outer product} \longrightarrow \text{inner product in the middle} \longrightarrow \text{number} \times \text{ket}}$$

You are going to use this *constantly*.

Quick Practice: Numbers and Operators

- Q1.** Is $\langle 0|1\rangle$ a number or an operator?
- Q2.** Is $|0\rangle \langle 1|$ a number or an operator?
- Q3.** Compute $|0\rangle \langle 1| |1\rangle$.
- Q4.** Compute $|0\rangle \langle 1| |0\rangle$.

Answers: Numbers and Operators

Q1. A **number**. Shapes: $(1 \times 2)(2 \times 1) = 1 \times 1$. Its value here is 0.

Q2. An **operator**. Shapes: $(2 \times 1)(1 \times 2) = 2 \times 2$, explicitly $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$.

Q3. Collapse the middle: $|0\rangle\langle 1|1\rangle = |0\rangle \cdot 1 = |0\rangle$.

Q4. Collapse the middle: $|0\rangle\langle 1|0\rangle = |0\rangle \cdot 0 = 0$.

Q3 and Q4 together are the whole personality of $|0\rangle\langle 1|$: it moves $|1\rangle$ to $|0\rangle$ and annihilates $|0\rangle$.

2.3 Projectors

Now take the special case where the bra and ket are the same:

$$|\psi\rangle\langle\psi|.$$

This gives a very important operator.

Projector

For a **unit vector** $|\psi\rangle$,

$$P_\psi = |\psi\rangle\langle\psi|$$

is the projector onto the direction $|\psi\rangle$.

It takes a state, extracts its $|\psi\rangle$ component, and keeps only that component.

What a Projector Does

Take an arbitrary state $|v\rangle$:

$$P_\psi |v\rangle = |\psi\rangle\langle\psi| |v\rangle = \langle\psi|v\rangle |\psi\rangle.$$

So the projector throws away everything except the part of $|v\rangle$ along $|\psi\rangle$.

Think:

$$\text{state} \longrightarrow \text{keep one direction}$$

It is literally the algebraic version of projecting a vector onto an axis.

The Two Basis Projectors

For the computational basis,

$$P_0 = |0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix},$$

and

$$P_1 = |1\rangle \langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Take

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle.$$

Then

$$P_0 |\psi\rangle = |0\rangle \langle 0| (\alpha |0\rangle + \beta |1\rangle) = \alpha |0\rangle,$$

while

$$P_1 |\psi\rangle = |1\rangle \langle 1| (\alpha |0\rangle + \beta |1\rangle) = \beta |1\rangle.$$

So the two projectors simply pick out the two components:

$$P_0 : |\psi\rangle \mapsto \alpha |0\rangle$$

$$P_1 : |\psi\rangle \mapsto \beta |1\rangle$$

Why $P^2 = P$

Projecting twice should not do anything extra.

Once you have removed everything except the $|\psi\rangle$ component, projecting again should just give the same thing.

Algebraically,

$$\begin{aligned} P_\psi^2 &= (|\psi\rangle \langle \psi|)(|\psi\rangle \langle \psi|) \\ &= |\psi\rangle \underbrace{\langle \psi | \psi \rangle}_{=1} \langle \psi| \\ &= |\psi\rangle \langle \psi| = P_\psi. \end{aligned}$$

Therefore,

$$P_\psi^2 = P_\psi$$

This property is called **idempotence**.

The State Must Be Normalised

The clean projector formula

$$P_\psi = |\psi\rangle \langle \psi|$$

assumes

$$\langle \psi | \psi \rangle = 1.$$

If $|\psi\rangle$ is not normalised, then

$$(|\psi\rangle \langle \psi|)^2 = \langle \psi | \psi \rangle |\psi\rangle \langle \psi|,$$

which is not equal to P_ψ unless $\langle \psi | \psi \rangle = 1$.

So before calling $|\psi\rangle \langle \psi|$ a projector, check the normalisation.

Quick Practice: Projectors

Q1. Write the projector onto $|0\rangle$.

Q2. Write the projector onto $|1\rangle$.

Q3. For $|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$, compute $P_0 |\psi\rangle$.

Q4. Show that $P_0^2 = P_0$.

Answers: Projectors

Q1. $P_0 = |0\rangle \langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$.

Q2. $P_1 = |1\rangle \langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$.

Q3. Collapse each term:

$$P_0 |\psi\rangle = |0\rangle (\underbrace{\alpha \langle 0|0\rangle}_1 + \underbrace{\beta \langle 0|1\rangle}_0) = \alpha |0\rangle.$$

Q4. Two routes, both fine.

Dirac: $P_0^2 = |0\rangle \langle 0|0\rangle \langle 0| = |0\rangle \langle 0| = P_0$.

Matrix: $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$.

The Dirac route is one line and generalises to any dimension. The matrix route works but stops being fun at 8×8 .

Completeness Appears

For the computational basis,

$$|0\rangle \langle 0| + |1\rangle \langle 1| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Therefore,

$$\boxed{|0\rangle \langle 0| + |1\rangle \langle 1| = I}$$

This is not a random coincidence.

We have just discovered the special case of the **completeness relation** that we will state properly in the next section.

3 The Completeness Relation

If $\{|i\rangle\}$ is an orthonormal basis, then

$$\sum_i |i\rangle \langle i| = I$$

This is the **completeness relation**, also called the **resolution of the identity**.

What Does It Mean?

The basis vectors contain all the directions needed to describe any state.
For any state $|\psi\rangle$,

$$|\psi\rangle = \sum_i |i\rangle \langle i|\psi\rangle$$

The inner product

$$\langle i|\psi\rangle$$

extracts the coefficient of $|i\rangle$, and the sum puts all the components back together.
So, in plain English:

break the state into basis components $\longrightarrow$ put them back together

That is all the completeness relation is saying.

For a Qubit

For the computational basis $\{|0\rangle, |1\rangle\}$,

$$|0\rangle \langle 0| + |1\rangle \langle 1| = I$$

Therefore, if

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle,$$

then

$$|\psi\rangle = |0\rangle \langle 0|\psi\rangle + |1\rangle \langle 1|\psi\rangle = \alpha |0\rangle + \beta |1\rangle.$$

The completeness relation is just the operator version of **expanding a vector in a basis**.

Orthonormal, Not Just Any Basis

The relation needs $\langle i|j\rangle = \delta_{ij}$. Unit length and mutually perpendicular.

Try it with $|0\rangle$ and $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, which span the space but are not orthogonal:

$$|0\rangle\langle 0| + |+\rangle\langle +| = \begin{pmatrix} 3/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix} \neq I.$$

The directions overlap, so the components get double-counted.

3.1 The Insert-Identity Trick

Sometimes we want to make the basis explicit inside an expression.

Since

$$I = \sum_i |i\rangle\langle i|,$$

we can simply insert it wherever we need:

$$A = IA = \left(\sum_i |i\rangle\langle i| \right) A.$$

Or on both sides,

$$A = IAI = \left(\sum_i |i\rangle\langle i| \right) A \left(\sum_j |j\rangle\langle j| \right).$$

This lets us turn an abstract operator into its matrix elements:

$$A = \sum_{i,j} |i\rangle\langle i| A |j\rangle\langle j|$$

or, since $\langle i| A |j\rangle$ is just a number,

$$A = \sum_{i,j} \langle i| A |j\rangle |i\rangle\langle j|$$

The Intuition

In ordinary algebra, multiplying by 1 changes nothing. Here we are doing the same thing, except we replace

$$1 \rightarrow I = \sum_i |i\rangle\langle i|$$

so that the **basis becomes visible**. That's the whole trick.

Pulling Out a Matrix Element

The entries of a matrix are obtained by sandwiching:

$$A_{ij} = \langle i| A |j\rangle$$

For $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$,

$$\langle 0|X|0\rangle = 0, \quad \langle 0|X|1\rangle = 1, \quad \langle 1|X|0\rangle = 1, \quad \langle 1|X|1\rangle = 0,$$

so

$$X = |0\rangle\langle 1| + |1\rangle\langle 0|$$

Read that out loud: *send $|1\rangle$ to $|0\rangle$, send $|0\rangle$ to $|1\rangle$* . The matrix tells you what X **is**. The outer-product form tells you what X **does**.

Quick Practice: Completeness and Insert-Identity

Q1. Complete the identity: $|0\rangle\langle 0| + |1\rangle\langle 1| = ?$

Q2. For $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, compute $\langle 0|\psi\rangle$.

Q3. Verify that $(|0\rangle\langle 0| + |1\rangle\langle 1|)|\psi\rangle = |\psi\rangle$.

Q4. Compute $\langle 0|Z|0\rangle$ and $\langle 0|Z|1\rangle$, and hence write Z in outer-product form.

Answers: Completeness and Insert-Identity

Q1. I .

Q2.

$$\langle 0|\psi\rangle = \alpha \underbrace{\langle 0|0\rangle}_1 + \beta \underbrace{\langle 0|1\rangle}_0 = \alpha.$$

The coefficient is not something you are handed. It is something you **extract**, with an inner product.

Q3.

$$|0\rangle \underbrace{\langle 0|\psi\rangle}_\alpha + |1\rangle \underbrace{\langle 1|\psi\rangle}_\beta = \alpha|0\rangle + \beta|1\rangle = |\psi\rangle. \quad \checkmark$$

Q4. $\langle 0|Z|0\rangle = 1$ and $\langle 0|Z|1\rangle = 0$. Finishing the other two gives $\langle 1|Z|0\rangle = 0$ and $\langle 1|Z|1\rangle = -1$, so

$$Z = |0\rangle\langle 0| - |1\rangle\langle 1|$$

In words: leave $|0\rangle$ alone, flip the sign on $|1\rangle$. Only projector terms appear, because Z is diagonal in this basis. Compare X , which is all off-diagonal terms, because all it does is move things around.

Summary of Part I

Ket is a column. Bra is the conjugate-transposed row. Bra-then-ket gives a number, ket-then-bra gives a matrix. Same-vector outer products are projectors. Projectors over a full orthonormal basis add up to I , and that fact is the most useful way to multiply by 1 anyone has invented.

Part II

Matrix Groups

4 What Is a Group?

Before we talk about matrix groups, we should say what the word **group** is doing there.

The Mental Model

A group is a **set of moves you are allowed to make**, plus a rule for combining them.

That's it.

group = things you can do + a way to do one after another

The key point, and the one people miss:

the things in a group do not have to be numbers.

They can be rotations, reflections, permutations, matrices, quantum gates, symmetries of a crystal. Anything you can combine and undo.

This is why groups are everywhere in physics. **Symmetries are transformations, and transformations can be combined.**

A Group You Already Know

Take the rotations of a square that leave it looking the same:

$$0^\circ, \quad 90^\circ, \quad 180^\circ, \quad 270^\circ.$$

Combine two by doing one and then the other. Four things are immediately true:

1. Doing two rotations gives another rotation from the list. (90° then 180° is 270° .)
2. Doing three in a row, the bracketing does not matter.
3. 0° is the “do nothing” move.
4. Every rotation can be undone. (90° is undone by 270° .)

Those four sentences *are* the group axioms. Nobody imposed them from outside. We read them off an example a five-year-old could handle.

4.1 The Four Rules

Now the formal version.

Group Axioms

A set G with an operation $\circ$ is a **group** if, for all $a, b, c \in G$:

1. Closure

$$a \circ b \in G.$$

Combining two elements gives another element of the group.

2. Associativity

$$(a \circ b) \circ c = a \circ (b \circ c).$$

The brackets can move.

3. Identity

There exists $e \in G$ with

$$e \circ a = a \circ e = a.$$

The identity does nothing.

4. Inverse

For every $a \in G$ there exists $a^{-1} \in G$ with

$$a \circ a^{-1} = a^{-1} \circ a = e.$$

The inverse undoes the move.

The Four Questions

When someone claims a set is a group, don't panic. Ask four questions:

Can I combine them?

Do I stay inside the set?

Is there a do-nothing element?

Can everything be undone?

Plus the quiet one that almost always comes free:

$$(ab)c = a(bc)$$

The axioms are just the formal version of those questions.

A Group Is a Set *Plus* an Operation

Never answer “is this a group?” without asking “under what operation?”

$\{0, 1\}$ is a group under XOR. Identity 0, and each element is its own inverse.

$\{0, 1\}$ is **not** a group under ordinary multiplication, because 0 has no inverse.

Same set. Different operation. Different answer.

Quick Practice: Groups

- Q1.** Consider rotations of the plane by multiples of 90° , combined by doing one then the other. What is the identity element?
- Q2.** In that same set, what is the inverse of the 90° rotation?
- Q3.** Is $\{1, -1\}$ closed under ordinary multiplication?
- Q4.** Is $\{0, 1, 2, 3, \dots\}$ a group under addition? If not, which axiom fails?

Solutions: Groups

Q1.

The identity is the move that does nothing:

$$0^\circ$$

because 0° followed by θ is just θ .

Q2.

We need the rotation that brings 90° back to the start:

$$90^\circ + 270^\circ = 360^\circ = 0^\circ.$$

Therefore

$$(90^\circ)^{-1} = 270^\circ$$

Q3.

Check every product:

$$(1)(1) = 1, \quad (1)(-1) = -1, \quad (-1)(1) = -1, \quad (-1)(-1) = 1.$$

Every answer is still in $\{1, -1\}$, so

Yes, it is closed.

(It is in fact a full group: identity 1, and each element is its own inverse.)

Q4.

No.

Closure holds, associativity holds, and 0 is a perfectly good identity. But **inverses fail**: there is no non-negative integer you can add to 3 to get back to 0.

Allow negatives and $\mathbb{Z}$ under addition is a group. One axiom is all it takes.

4.2 Now Replace the Moves with Matrices

The Bridge

A matrix *is* a transformation. So instead of saying

“combine two rotations”

we say

“multiply two matrices”.

The dictionary is short:

Group language	Matrix language
$a \circ b$	AB
identity e	I
inverse a^{-1}	A^{-1}

Nothing new has happened. We swapped abstract “moves” for matrices, and the four questions still work.

So when the next page says

$U(n) = \{U : U^\dagger U = I\}$ is a group under matrix multiplication,

you already know exactly what is being claimed: you can multiply two of them and stay inside, I is in there, and everything has an inverse that is also in there.

Why Bother

Group theory is what stops quantum gates from being a random pile of matrices you have to memorise. It tells you which ones are related, which ones generate the others, and which combinations are even possible.

5 Unitary Operators

Quantum gates are not arbitrary matrices. They are **unitary**.

Unitary

U is unitary if

$$U^\dagger U = U U^\dagger = I$$

equivalently

$$U^{-1} = U^\dagger$$

So inverting a unitary costs nothing. Conjugate, transpose, done.

Why We Care

Unitary operators preserve inner products:

$$\langle Uu|Uv \rangle = \langle u|v \rangle,$$

and therefore preserve length:

$$\|U|\psi\rangle\| = \||\psi\rangle\|.$$

So if you start with a normalised state, you stay normalised.

unitary = rotates the state, never stretches or shrinks it

This is also why gates are reversible, which was the whole opening argument of Course Pack 01. Nothing gets destroyed, so nothing needs to be forgotten.

Quick Practice: Unitary Operators

- Q1.** Check that X is unitary by computing $X^\dagger X$.
- Q2.** If U is unitary, what is U^{-1} ?
- Q3.** If $\langle\psi|\psi\rangle = 1$ and U is unitary, what is $\langle\psi|U^\dagger U|\psi\rangle$?
- Q4.** Is the projector $|0\rangle\langle 0|$ unitary? Justify in one line.

Answers: Unitary Operators

- Q1.** X is real and symmetric, so $X^\dagger = X$, and $X^\dagger X = X^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$. Unitary. ✓
- Q2.** $U^{-1} = U^\dagger$. This is the entire selling point of unitarity.
- Q3.** $\langle\psi|U^\dagger U|\psi\rangle = \langle\psi|I|\psi\rangle = \langle\psi|\psi\rangle = 1$. Normalisation survives.
- Q4. No.** $(|0\rangle\langle 0|)^\dagger |0\rangle\langle 0| = |0\rangle\langle 0| \neq I$. It cannot be unitary anyway, because it kills $|1\rangle$ entirely, and unitaries are invertible. Projectors throw information away. That is exactly what they are for, and exactly why they are not gates.

6 The Groups $U(n)$ and $SU(n)$

Course Pack 01 showed that the unitaries form a group:

$$U(n) = \{U \in \mathbb{C}^{n \times n} : U^\dagger U = I\}$$

Now we ask what sits inside it.

Determinants of Unitaries Are Stuck on the Circle

Take determinants of $U^\dagger U = I$, using $\det(A^\dagger) = \overline{\det A}$:

$$1 = \det(U^\dagger U) = \overline{\det U} \det U = |\det U|^2.$$

So

$$|\det U| = 1$$

The determinant of a unitary is never just any complex number. It always lands somewhere on the unit circle, which is the group $U(1)$ from Course Pack 01.

Since every unitary determinant lands on the circle, the obvious thing to do is single out the ones that land exactly on 1.

Special Unitary Group

$$SU(n) = \{ U \in U(n) : \det U = 1 \}$$

Special always means “determinant one”. Nothing more exotic than that.

It Is a Subgroup, and the Check Is Four Lines

Closure: $\det(AB) = \det A \det B = 1 \cdot 1 = 1$.

Identity: $\det I = 1$.

Inverse: $\det(A^{-1}) = 1/\det A = 1$.

Associativity: inherited from matrix multiplication, as always.

Every step used only one fact: the determinant turns products into products. That is why this works and why it works for every n .

Quick Practice: $U(n)$ and $SU(n)$

Q1. Is I in $SU(2)$?

Q2. Compute $\det X$. Is X in $SU(2)$?

Q3. If $\det A = 1$ and $\det B = 1$, what is $\det(AB)$?

Q4. If $A \in SU(2)$, what is $\det(A^{-1})$?

Answers: $U(n)$ and $SU(n)$

Q1. Yes. I is unitary and $\det I = 1$.

Q2. $\det X = (0)(0) - (1)(1) = -1$. So **no**, X is not in $SU(2)$. It is unitary, and $|-1| = 1$ as required, but it is not 1.

Q3. 1. This is closure.

Q4. 1. This is the inverse axiom. Note $A^{-1} = A^\dagger$ here, so you could also say $\det(A^\dagger) = \overline{\det A} = \bar{1} = 1$.

7 Sorting the Gates

One determinant, one answer.

Gate	Matrix	det	In $SU(2)$?
I	$\text{diag}(1, 1)$	1	yes
X	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	-1	no
Y	$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$	-1	no
Z	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	-1	no
H	$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	-1	no
S	$\text{diag}(1, i)$	i	no
T	$\text{diag}(1, e^{i\pi/4})$	$e^{i\pi/4}$	no
$R_z(\theta)$	$\text{diag}(e^{-i\theta/2}, e^{i\theta/2})$	1	yes

Watch the Prefactor on H

For a 2×2 matrix,

$$\det(cA) = c^2 \det A$$

so the $\frac{1}{\sqrt{2}}$ in front of H contributes $\frac{1}{2}$, not $\frac{1}{\sqrt{2}}$:

$$\det H = \frac{1}{2}[(1)(-1) - (1)(1)] = \frac{1}{2}(-2) = -1.$$

Forgetting to square the prefactor is the standard way to lose this mark.

The Rotation Gates Are the $SU(2)$ Ones

$$R_z(\theta) = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}, \quad \det R_z(\theta) = e^{-i\theta/2} e^{i\theta/2} = e^0 = 1.$$

Same for $R_x(\theta)$ and $R_y(\theta)$. The rotation gates were built to sit in $SU(2)$. The named gates X, Y, Z, H, S, T were not.

A Question We Are Not Answering Today

Notice

$$R_z(\pi) = \text{diag}(-i, i) = -i Z.$$

So Z and $R_z(\pi)$ differ by an overall factor of $-i$, which has modulus 1. Do two matrices differing by such a factor describe the *same physical operation*? Real question, real answer, and it needs the measurement rule, which we do not have yet. For today this is algebra and nothing more. Park it.

Quick Practice: Sorting the Gates

- Q1.** Compute $\det S$.
- Q2.** Compute $\det H$. Careful with the prefactor.
- Q3.** Compute $\det R_z(\theta)$.
- Q4.** Which of Z and $R_z(\pi)$ is in $SU(2)$?

Answers: Sorting the Gates

- Q1.** Diagonal, so multiply the entries: $\det S = 1 \cdot i = i$. Not in $SU(2)$.
- Q2.** $\det H = \frac{1}{2}[(1)(-1) - (1)(1)] = -1$. If you got $-\frac{1}{\sqrt{2}}$ or $-\sqrt{2}$, you did not square the prefactor.
- Q3.** $e^{-i\theta/2} \cdot e^{i\theta/2} = 1$, for every θ .
- Q4.** $R_z(\pi)$ is in $SU(2)$; Z is not, since $\det Z = -1$. They differ only by the factor $-i$, and that factor is exactly what the determinant condition is picking up on.

8 What $SU(2)$ Actually Looks Like

$SU(2)$ is small enough to write down *completely*. That is rare, so enjoy it.

Every Element of $SU(2)$

$$U = \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix}, \quad |\alpha|^2 + |\beta|^2 = 1$$

with $\alpha, \beta \in \mathbb{C}$.

Every matrix of this form is in $SU(2)$, and every element of $SU(2)$ has this form. Two complex numbers and one constraint, and you have the entire group.

Checking It

Determinant:

$$\det U = \alpha\bar{\alpha} - (-\bar{\beta})\beta = |\alpha|^2 + |\beta|^2 = 1. \quad \checkmark$$

Unitarity:

$$U^\dagger U = \begin{pmatrix} \bar{\alpha} & \bar{\beta} \\ -\beta & \alpha \end{pmatrix} \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix} = \begin{pmatrix} |\alpha|^2 + |\beta|^2 & 0 \\ 0 & |\alpha|^2 + |\beta|^2 \end{pmatrix} = I. \quad \checkmark$$

The off-diagonal entries cancel on their own. The diagonal ones need the normalisation.

Counting the Freedom

Two complex numbers is **four** real numbers.

One real equation, $|\alpha|^2 + |\beta|^2 = 1$, removes **one**.

$SU(2)$ has 3 real parameters

$U(2)$ has 4

The extra parameter in $U(2)$ is exactly the determinant.

Three is also how many numbers you need to specify a rotation of ordinary 3D space. That is not a coincidence, and it is why R_x, R_y, R_z are enough to build any single-qubit gate up to an overall factor.

Quick Practice: The Shape of $SU(2)$

Q1. Write the $SU(2)$ matrix for $\alpha = 1, \beta = 0$.

Q2. Write the $SU(2)$ matrix for $\alpha = 0, \beta = 1$.

Q3. Find α and β for $R_z(\theta)$.

Q4. How many real parameters does $SU(2)$ have, and where did the missing one go?

Answers: The Shape of $SU(2)$

Q1. $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$.

Q2. $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Check: $|0|^2 + |1|^2 = 1$. $\checkmark$ Bonus: this equals $-iY$, so it is Y dragged into $SU(2)$ by a phase.

Q3. $R_z(\theta) = \text{diag}(e^{-i\theta/2}, e^{i\theta/2})$ is diagonal, so $\beta = 0$ and $\alpha = e^{-i\theta/2}$. The form then demands the bottom-right entry be $\bar{\alpha} = e^{i\theta/2}$, which matches. $\checkmark$

Worth noticing: $\beta = 0$ forces U diagonal, and then α is one number of modulus 1. So the diagonal part of $SU(2)$ is a one-parameter family, and that family is exactly $\{R_z(\theta)\}$. That is why R_z has one angle and not three.

Q4. Three. Four real numbers in (α, β) , minus one real constraint from $|\alpha|^2 + |\beta|^2 = 1$.

Where This Goes Next, One Line Each

Generators. Every unitary can be written $U = e^{iA}$ with A Hermitian, and for $SU(2)$ the allowed A are spanned by X, Y, Z . So the Pauli matrices *generate* $SU(2)$.

Bloch sphere. Those three parameters become an axis and an angle, and every qubit state sits on a sphere.

Both are Unit 2. Both are the reason we bothered with $SU(2)$ today.

Fortune Cookie

“Every group in this course is either a way of shuffling things or a way of rotating them. $SU(2)$ is the second kind, wearing a disguise.”

Shunya

9 Problem Set

The Quick Practice blocks tested one thing each. These mix them.

Problems for Next Class

1. Compute all four outer products $|0\rangle\langle 0|$, $|0\rangle\langle 1|$, $|1\rangle\langle 0|$, $|1\rangle\langle 1|$ as explicit matrices. Which of them are projectors?
2. Let $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$. Show $\{|+\rangle, |-\rangle\}$ is orthonormal, and verify $|+\rangle\langle +| + |-\rangle\langle -| = I$.
3. Using $X = |0\rangle\langle 1| + |1\rangle\langle 0|$ and the collapse pattern only, with no matrices, compute $X|+\rangle$ and $X|-\rangle$. What have you just found?
4. Using outer products and the collapse pattern only, prove $X^2 = I$.
5. Compute XZ and ZX as matrices. Is $XZ = ZX$? Write each in the form $\pm iP$ for a Pauli P .
6. Compute $\det$ for each of X, Y, Z, H, S, T . Which are in $SU(2)$? Check every answer has modulus 1, as §6 requires.
7. Show $SU(2)$ is a subgroup of $U(2)$, checking all four axioms, using only $\det(AB) = \det A \det B$ and $\det(A^{-1}) = 1/\det A$.
8. Compute $\langle +|Z|+\rangle$, $\langle +|Z|-\rangle$, $\langle -|Z|+\rangle$, $\langle -|Z|-\rangle$, and write the matrix of Z in the ordered basis $\{|+\rangle, |-\rangle\}$. Compare it with the matrix of X in the computational basis.

9.1 Solutions

P1 — The four outer products

$$|0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad |0\rangle\langle 1| = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad |1\rangle\langle 0| = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad |1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Each has a single 1, sitting in position (i, j) for $|i\rangle\langle j|$.

Which are projectors? Only the two with matching labels:

$$(|0\rangle\langle 0|)^2 = |0\rangle\langle 0| \underbrace{|0\rangle\langle 0|}_{1} = |0\rangle\langle 0|, \quad (|0\rangle\langle 1|)^2 = |0\rangle\langle 1| \underbrace{|1\rangle\langle 0|}_{0} = 0.$$

So $|0\rangle\langle 0|$ and $|1\rangle\langle 1|$ are projectors. The mismatched ones square to zero.

P2 — Completeness in the $\{|+\rangle, |-\rangle\}$ basis

Orthonormality.

$$\langle +|+\rangle = \frac{1}{2}(1 + 0 + 0 + 1) = 1, \quad \langle +|-\rangle = \frac{1}{2}(1 - 0 + 0 - 1) = 0.$$

Completeness. Expand both:

$$|+\rangle \langle +| = \frac{1}{2}(|0\rangle \langle 0| + |0\rangle \langle 1| + |1\rangle \langle 0| + |1\rangle \langle 1|),$$

$$|-\rangle \langle -| = \frac{1}{2}(|0\rangle \langle 0| - |0\rangle \langle 1| - |1\rangle \langle 0| + |1\rangle \langle 1|).$$

Add. The cross terms cancel, the diagonal terms double:

$$|+\rangle \langle +| + |-\rangle \langle -| = |0\rangle \langle 0| + |1\rangle \langle 1| = I.$$

The point. Completeness is not special to the computational basis. It holds in *every* orthonormal basis, which is what makes insert-identity usable anywhere.

P3 — X acting on $|\pm\rangle$

$$X|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle \langle 1| + |1\rangle \langle 0|)(|0\rangle + |1\rangle).$$

Four terms. Two die because $\langle 1|0\rangle = 0$ and $\langle 0|1\rangle = 0$, leaving

$$X|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle.$$

Same computation on $|-\rangle$:

$$X|-\rangle = \frac{1}{\sqrt{2}}(-|0\rangle + |1\rangle) = -|-\rangle.$$

What you found. $|+\rangle$ and $|-\rangle$ are the eigenvectors of X , with eigenvalues $+1$ and -1 . No characteristic polynomial, no row reduction. This is the single best advertisement for the notation.

P4 — $X^2 = I$ with no matrices

$$X^2 = (|0\rangle \langle 1| + |1\rangle \langle 0|)(|0\rangle \langle 1| + |1\rangle \langle 0|).$$

Expand into four terms. The two inner symbols meet in each, and only matching ones survive:

$$\underbrace{|0\rangle \langle 1|0\rangle \langle 1|}_0 + \underbrace{|0\rangle \langle 1|1\rangle \langle 0|}_{|0\rangle \langle 0|} + \underbrace{|1\rangle \langle 0|0\rangle \langle 1|}_{|1\rangle \langle 1|} + \underbrace{|1\rangle \langle 0|1\rangle \langle 0|}_0.$$

So

$$X^2 = |0\rangle \langle 0| + |1\rangle \langle 1| = I,$$

the last step by completeness.

Note. The final step was not a calculation. It was *recognising* completeness when it turned up. Without that, you stop one line early with a sum of projectors and think you are not done.

P5 — XZ versus ZX

$$XZ = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad ZX = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Every entry flipped sign, so

$$\boxed{XZ = -ZX}$$

These matrices **anticommute**. The answer to “is $XZ = ZX$ ” is a firm no.

In Pauli form. Since $-iY = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$,

$$XZ = -iY, \quad ZX = +iY.$$

Two things worth noticing. First, order matters, which is the non-abelian point from Course Pack 01 showing up as plain arithmetic. Second, $-iY$ is not in $\{I, X, Y, Z\}$, which is precisely why that set failed closure and needed the phases $\{\pm 1, \pm i\}$ bolted on to become a group.

P6 — Determinants of the standard gates

$$\det X = -1, \quad \det Y = (0)(0) - (-i)(i) = -1, \quad \det Z = -1, \\ \det H = \frac{1}{2}[(1)(-1) - (1)(1)] = -1, \quad \det S = i, \quad \det T = e^{i\pi/4}.$$

Answer: none of the six is in $SU(2)$.

Modulus check. $|-1| = 1$, $|i| = 1$, $|e^{i\pi/4}| = 1$. All six land on the unit circle, exactly as required. If any of your answers has modulus other than 1, you made an arithmetic error, and this is the fastest way to catch it.

Usual slips. Forgetting the $\frac{1}{2}$ on H , and mishandling Y : note $(-i)(i) = -i^2 = 1$, so subtracting it gives -1 .

P7 — $SU(2) \leq U(2)$

Elements of $SU(2)$ are unitary by definition, so only the determinant needs tracking.

Closure. $\det(AB) = \det A \det B = 1$, and AB is unitary. So $AB \in SU(2)$.

Associativity. Inherited from matrix multiplication.

Identity. I is unitary and $\det I = 1$.

Inverse. $A^{-1} = A^\dagger$ exists and is unitary, and $\det(A^{-1}) = 1/\det A = 1$.

Remark. Nothing in that used the number 2. The same four lines prove $SU(n) \leq U(n)$ for every n .

P8 — Z in the $\{|+\rangle, |-\rangle\}$ basis

First act with $Z = |0\rangle\langle 0| - |1\rangle\langle 1|$:

$$Z|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle, \quad Z|-\rangle = |+\rangle.$$

Then take inner products:

$$\langle +|Z|+\rangle = \langle +|-\rangle = 0, \quad \langle +|Z|-\rangle = \langle ++\rangle = 1, \quad \langle -|Z|+\rangle = 1, \quad \langle -|Z|-\rangle = 0.$$

So in the ordered basis $\{|+\rangle, |-\rangle\}$,

$$[Z] = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The punchline. That array is the matrix of X in the computational basis. Z and X are still different operators. What happened is that Z *written in the Hadamard basis* looks exactly like X *written in the computational basis*. This is $HZH = X$ from Course Pack 01, seen from the other side.

Moral. A matrix without a stated basis is incomplete information. Course Pack 01 asserted this. Here it bites.

10 Reading and What Comes Next

Reading

- Nielsen & Chuang §2.1.2–2.1.4. Dirac notation, inner and outer products, completeness. Short.
- Nielsen & Chuang §4.2, for R_x, R_y, R_z and where they sit in $SU(2)$. Skim only.

Next class

The postulates of quantum mechanics. States as unit vectors, evolution as unitary maps, observables as Hermitian operators, and the Born rule.

The projectors and the completeness relation from today are the machinery the measurement postulate is written in, and the overall factor we parked in §7 finally gets settled.

Quantum Entanglement & Quantum Computing PES University

Do the Quick Practice blocks. They are four questions each and they are the point.